

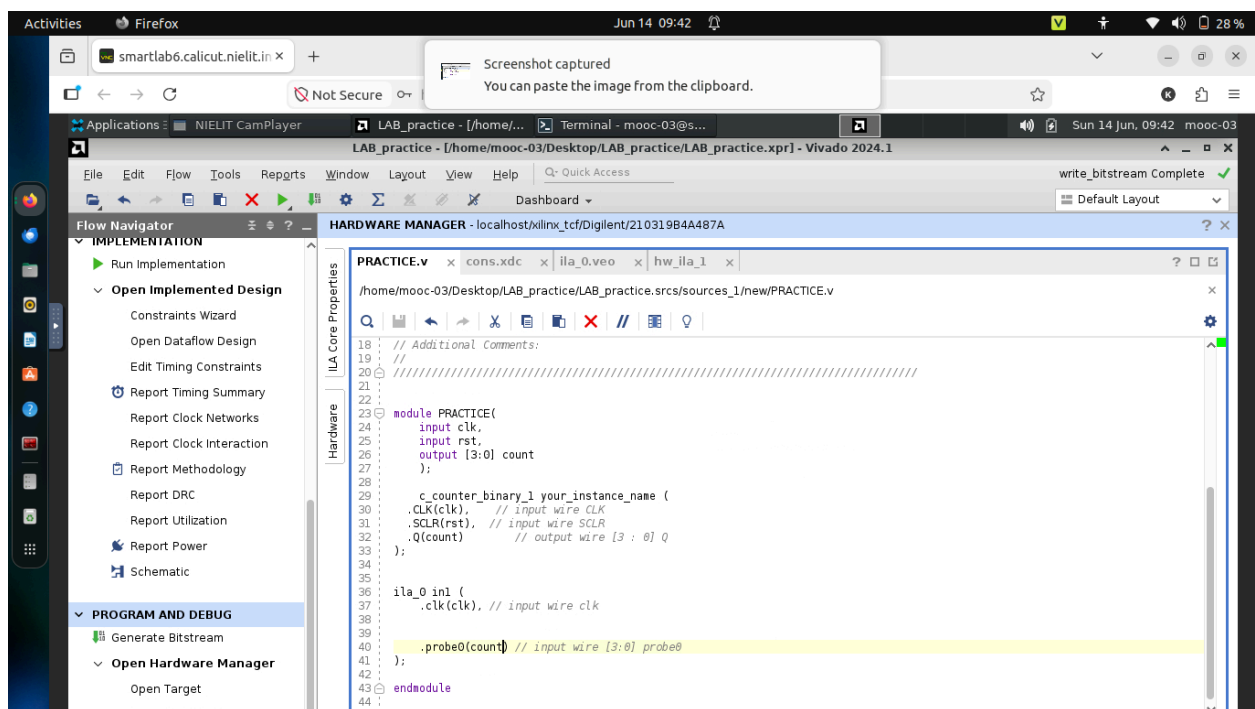
FPGA Synthesis and Prototyping Lab:

39

Counters

Introduction

The purpose of this experiment is to Synthesis and Check the behavior of the Counters mentioned in Lab 18,19,20,21



Activities Firefox Jun 14 09:42 28 %

smartlab6.calicut.nielit.in x

Not Secure http://smart.calicut.nielit.in:13595/vnc.html

Applications: NIELIT CamPlayer LAB_practice - [/home/... Terminal - mooc-03@... Sun 14 Jun, 09:42 mooc-03

LAB_practice - [/home/mooc-03/Desktop/LAB_practice/LAB_practice.xpr] - Vivado 2024.1

File Edit Flow Tools Reports Window Layout View Help Q-Quick Access write_bitstream Complete ✓ Default Layout

Flow Navigator

- Create Block Design
- Open Block Design
- Generate Block Design
- SIMULATION**
 - Run Simulation
- RTL ANALYSIS**
 - Run Linter
 - Open Elaborated Design
 - Report Methodology
 - Report DRC
 - Report Noise
 - Schematic
- SYNTHESIS**
 - Run Synthesis
 - Open Synthesized Design
 - Constraints Wizard
 - Edit Timing Constraints

ELABORATED DESIGN - xc7a100tcsg324-1

Sources Netlist x

- PRACTICE
 - Nets (6)
 - in1 (ila_0)
 - your_instance_name (c_counter_binary_1)

Properties

Select an object to see properties

Project Summary x Schematic x PRACTICE.v x cons_1 x

2 Cells 6 I/O Ports 6 Nets

Schematic

clk your_instance_name in1
CLK SCLR Q(3:0) probeQ(3:0) ila_0
rst c_counter_binary_1 count[3:0]

Tcl Console Messages Log Reports Design Runs x

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed
✓ synth_1 (active)	constrs_1	synth_design Complete!								
✓ impl_1	constrs_1	write_bitstream Complete!	26.821	0.000	0.030	0.000		0.000	0.101	
Out-of-Context Module Runs										

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Flow Navigator

- Run Implementation
- Open Implemented Design**
 - Constraints Wizard
 - Open Dataflow Design
 - Edit Timing Constraints
 - Report Timing Summary
 - Report Clock Networks
 - Report Clock Interaction
 - Report Methodology
 - Report DRC
 - Report Utilization
 - Report Power
 - Schematic
- PROGRAM AND DEBUG**
 - Generate Bitstream
 - Open Hardware Manager
 - Open Target
 - Program Device

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210319B4A487A

PRACTICE.v x cons.xdc x ila_0.veo x hw_ila_1 x

/home/mooc-03/Desktop/LAB_practice/LAB_practice.srscs/constrs_1/new/cons.xdc

```
1 set_property PACKAGE_PIN A8 [get_ports rst]
2 set_property IOSTANDARD LVCMOS33 [get_ports rst]
3 set_property IOSTANDARD LVCMOS33 [get_ports clk]
4 set_property IOSTANDARD LVCMOS33 [get_ports {count[3]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {count[2]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {count[1]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {count[0]}]
8 set_property PACKAGE_PIN H5 [get_ports {count[3]}]
9 set_property PACKAGE_PIN J5 [get_ports {count[2]}]
10 set_property PACKAGE_PIN T9 [get_ports {count[1]}]
11 set_property PACKAGE_PIN T10 [get_ports {count[0]}]
12 set_property PACKAGE_PIN E3 [get_ports clk]
13
```

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smartlab6.calicut.nielit.in x +

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Applications : NIELIT CamPlayer LAB_practice - [/home/... Terminal - mooc-03@... Sun 14 Jun, 09:48 mooc-03

LAB_practice - [/home/mooc-03/Desktop/LAB_practice/LAB_practice.xpr] - Vivado 2024.1

File Edit Flow Tools Reports Window Layout View Help Q- Quick Access write_bitstream Complete ✓ Default Layout

Flow Navigator

- Report Power
- Schematic
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design
 - Constraints Wizard
 - Open Dataflow Design
 - Edit Timing Constraints
 - Report Timing Summary
 - Report Clock Networks
 - Report Clock Interaction
 - Report Methodology
 - Report DRC
 - Report Utilization
 - Report Power
 - Schematic
- PROGRAM AND DEBUG

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/21031984A487A

PRACTICE.v x cons.xdc x ila_0.veo x hw_ila_1 x

Waveform - hw_ila_1

ILA Status: Idle

Name	Value
count_OBUF[3:0]	c

Updated at: 2026-Jun-14 09:41:21

Settings - hw_ila_1 Status - hw_ila_1

Core status: Idle

Capture status - Window 1 of 1

Window sample 0 of 1024

Trigger Setup - hw_ila_1 Capture Setup - hw_ila_1

Name	Operator	Radix	Value
count_OBUF[3:0]	--	[H]	X

